

Lane A. Lindstrom

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EDUCATION

Brigham Young University | Provo, UT

Junior, Expected Dec 2027

B.S. Mathematics — Applied and Computational Mathematics Emphasis (ML & Data Science); Minor in Computer Science

- 3.95 University GPA
- BYU Scholarship (3 years, half tuition); Utah Opportunity Scholarship (2 years, \$1,000/semester); Marc & Shirley Burton Scholarship (1 year, \$1,000/semester)

SKILLS

- **Languages:** Python, TypeScript, SQL, C++, Bash
- **ML / Data Science:** NumPy, SciPy, Matplotlib
- **Backend / Systems:** FastAPI, asyncio, pytest, dataclasses, httpx, Websockets
- **Frontend / Frameworks:** React, Node.js, HTML, CSS
- **Tools & Platforms:** Git, GitHub, Docker, Linux (Arch, Ubuntu), Windows, Vercel
- **Mathematics:** Fourier & Haar Wavelet Transform, Linear Algebra, Complex Analysis, Markov Chains, Bayesian Statistics, Monte Carlo Methods, Algorithm Analysis (Temporal and Spatial Complexity), Linear & Circular Convolution

PROJECTS

Takes A Village — Multiplayer Browser Game + Evolutionary AI

- Designed evolutionary AI agents with genetic algorithms, combining goal-oriented and utility-based agent architectures
- Built data visualizations tracking AI training progress and fitness, with over 1,640 training games and over 16,000 genomes
- 4 Docker containers for a database, bot server, backend, and frontend with 122 automated tests
- Over 15,600 lines of python, and over 6,300 lines of TypeScript over nearly 400 commits in 3 months with a collaborator

Facial Recognition — Linear Algebra and Applied SVD Decomposition

- Implemented facial recognition using Principal Component Analysis (Eigenfaces) at 92.4% accuracy using s=25 eigenfaces

Fourier Transform and Signal Processing

- FFT implementation speed increased the speed 5x for n=256, 20x for n=1024, and 122x for n=4096 when compared to DFT
- Built an FFT-based linear/circular audio convolution engine and frequency-domain noise filter; removed corrupting factors from a license plate image using few frequency components to allow for clearer image analysis, and eliminated 97-100% of noise-band energy from corrupted audio recordings while preserving underlying signal content

EXPERIENCE

Student Research Assistant, Computer Science | Brigham Young University — Provo, UT

May 2026 – Current

- Worked in close collaboration with another student to produce the web browser game Takes A Village in 3 months
- Gave structure to the frontend using a Model View Presenter architecture
- Used dataclasses and reusable command class architecture to simplify the backend and use of websockets
- Created GOAP and Utility Evolutionary AI bots both with over 50 genes, allowing for complex decision making

Calculus I and Linear Algebra Grader | Brigham Young University — Provo, UT

Sept 2025 – Apr 2026

- Graded 2–3 written assignments weekly for combined course sections of 257 and 248 students, working with 9 other graders
- Identified errors in logic and reasoning to provide constructive, individualized feedback, and correct problem solving techniques

Student Research Assistant, Mathematics | Brigham Young University — Provo, UT

Sept 2025 – Dec 2025

- Collaborated with a team applying the circle method to derive an asymptotic representation for a restricted partition generating function, proving Rademacher's Theorem could not apply to our application, and driving focus to Meinardus' Theorem
- Rephrased, simplified, and helped prove/disprove prior research contributions for clarity and rigor

Public Representative | Church of Jesus Christ of Latter-day Saints — Argentina

Aug 2022 – Aug 2024

- Supervised and trained 75+ fellow service members over the course of a year
- Organized and led community service projects, fostering teamwork among participants planning and leading activities weekly
- Facilitated outreach programs to strengthen community engagement; fluent in Spanish